A Real-time VLM Framework for Hazard Inference in Home Environment for Mixed Reality Systems

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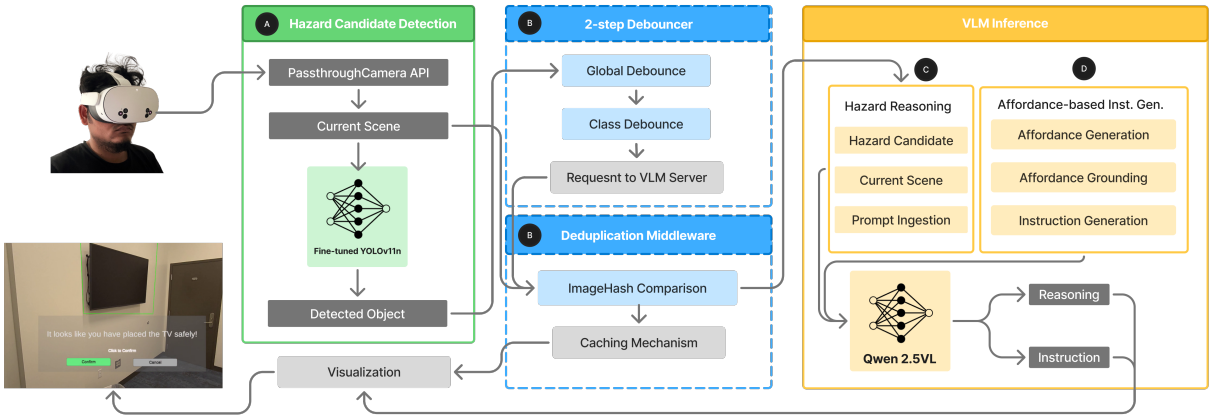


Figure 1: **Overview of the proposed framework for real-time hazard infrence in XR environments.** (a) Candidate hazard objects are first detected using a fine-tuned YOLOv11n model running locally on the HMD. When a candidate object is detected, the current scene and detection information are prepared for VLM-based reasoning. (b) Before transmission to the VLM server, two optimization strategies are applied: a two-step debouncing mechanism that reduces redundant server calls, and a deduplication middleware that prevents visually similar frames from being repeatedly processed. (c) The VLM then uses the scene context and detection information to infer whether the detected object represents a potential hazard. (d) If a hazard is confirmed, the system generates an affordance-based safety instruction to guide the user toward an appropriate corrective action.

Abstract

Unintentional injuries are a leading cause of child mortality, with many occurring in home environments. Mixed Reality (MR) systems could support household hazard awareness by providing contextual information directly within the user's physical environment. However, integrating Vision-Language Models (VLMs) into real-time MR applications is challenging because of their inference latency, computational cost, and the limited resources of head-mounted devices. This paper presents an optimized Object Detection and VLM-based framework for real-time household hazard identification and safety guidance in MR. The framework combines lightweight object detection for continuously identifying predefined hazard candidates with VLM-based reasoning for evaluating hazards in their surrounding context. To support practical real-time operation, the system incorporates request filtering, scene deduplication, class-based debouncing mechanisms, and precomputed affordance-based safety guidance to reduce redundant VLM inference and computational overhead. The framework is demonstrated through an MR-based parent-training prototype addressing common household hazards. Experimental results show an average VLM inference latency of 0.987 seconds while the object detection and MR application operate at approximately 67 FPS. These results demonstrate the technical feasibility of integrating optimized VLM-based contextual reasoning into real-time MR hazard-awareness systems.

Index Terms: Child Safety, Home Safety, Parent Education, Injury Prevention, Real-time VLM.

1 Introduction

Unintentional injuries remain a leading cause of child mortality, with many incidents occurring within home environments [2]. Common household hazards include unsafe sleep environments, unstable televisions, accessible cleaning products, and unsecured firearms [13, 2, 21]. To mitigate these risks, child safety education has traditionally relied on public health campaigns, printed materials, web-based resources, and caregiver training programs that promote safe sleep practices, hazard prevention, and injury reduction [13, 6, 3]. Although these approaches can improve caregiver awareness, they provide limited real-time, personalized, and context-aware feedback within the caregiver's own living environment.

Mixed Reality (MR) and Extended Reality (XR) technologies offer new opportunities for immersive and experiential safety training by allowing users to interact directly with their physical surroundings while receiving contextual guidance [20, 4, 17, 8, 14]. Prior systems have explored real-time guidance through visual augmentation and affordance-based interaction [10], suggesting the potential of MR to support environment-aware child safety training. Similarly, computer vision approaches such as CribNet [22] have demonstrated the feasibility of vision-based child hazard detection,

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although these methods are often limited to specific environments or predefined hazard categories.

A key challenge for MR-based hazard awareness systems is reliable hazard understanding in unconstrained home environments. Object detection models are effective for recognizing predefined object categories in real time, while Vision-Language Models (VLMs) provide broader semantic understanding and contextual reasoning capabilities for assessing safety situations [7]. However, deploying VLM-based reasoning in real-time MR applications remains challenging due to inference latency and the computational constraints of mobile and head-mounted devices [11, 16, 19, 9].

To address these challenges, we present an optimized real-time MR framework that integrates lightweight Object Detection with VLM-based contextual reasoning for household hazard identification and safety guidance. This work focuses on the technical integration and optimization required to make contextual VLM reasoning feasible within an interactive MR system, laying the groundwork for future evaluations of a complete parent-training intervention. Our primary contributions are threefold:

1. A hybrid hazard detection framework that combines object detection with VLM-based contextual hazard reasoning for identifying household safety risks.

2. An affordance-driven, human-centered guidance mechanism that generates context-aware safety recommendations based on scene understanding, preventing generic instructions.

3. An optimized real-time VLM deployment pipeline incorporating request filtering and scene deduplication strategies to reduce latency and computational overhead in MR environments.

2 Real-time Hazard Inference Framework

In this section, we describe the overall framework of the proposed system. The framework consists of three main components: an object detection module for identifying candidate objects in the scene (Figure 1 (a)), a VLM-based hazard reasoning module for contextual hazard identification and instruction generation (Figure 1 (c,d)), and a set of optimization strategies designed to reduce the computational load on the VLM server (Figure 1 (b)). These mechanisms enable efficient processing and support real-time performance suitable for the real-time hazard identification and training deployed in MR.

2.1 Hazard Candidate Detection

For the current child-safety training prototype, we focus on four common categories of household hazards that are frequently associated with infant and child injuries (Tab. 1), illustrated in Figure 2. Although recent studies such as HomeSafeBench [7] have demonstrated the potential of VLMs for safety reasoning in embodied environments, they also highlight limitations in reliably identifying safety hazards directly from complex household scenes. In particular, standalone VLM-based systems may struggle with real-time operation, even smaller VLM are not fast enough for edge devices with lower memories. Therefore, instead of relying solely on a VLM to identify hazards, we adopt a two-stage pipeline where candidate objects are first detected using a lightweight object detection model. Specifically, we train an object detection model to identify the following candidate objects in the environment, shown in Tab. 1. We fine-tune a YOLOv11n detector using a custom dataset collected from multiple sources. We combined images with labels from Objects365 [18], Google Open Images [5], LFC [12] and publicly available images from Roboflow. The trained detector is deployed directly on the HMD using the Unity Sentis.



Figure 2: Examples of household hazard scenarios considered in this work: (top-left) toys placed inside a baby crib, (top-right) unsafe television placement, (bottom-left) cleaning products accessible to children, and (bottom-right) an unsecured firearm.

2.2 VLM-Based Hazard Reasoning

Once a candidate object is detected, the corresponding image frame and object context are transmitted to a VLM server for further hazard analysis. Due to the computational limitations of current HMDs, the VLM inference is performed on a separate server. We used Qwen2.5-VL-7B-Instruct [1], an open-source vision language model, in our work. VLM analyzes the surrounding visual context to determine whether the detected object represents a safety hazard. To achieve this, we design category-specific prompts that guide the VLM to focus on particular safety conditions relevant to each candidate object. These prompts provide clear instructions regarding how the image should be analyzed and constrain the output format to ensure consistent responses.

2.3 Optimization Strategies

To enable real-time operation of the MR system while incorporating VLM inference, several optimization strategies were introduced.

2.3.1 Two-Step Debouncing for Request Control

When a potential hazard candidate object is detected, the current scene texture along with the detected object class is forwarded to the VLM server for contextual hazard reasoning. Because object detection runs at real-time frame rates, it may generate multiple detections for a repetitive object within a very short time window, potentially resulting in a large number of redundant VLM requests. To mitigate this issue, we introduce a two-step debouncing mechanism that regulates request generation. The first component is a *global debouncer*, which prevents the system from sending VLM requests more frequently than a predefined interval (500 ms in our implementation). This ensures that the system does not overwhelm the server with rapid consecutive requests. However, relying solely on a global debouncer may suppress legitimate hazard detections if a different object class appears within the debounce window. Therefore, we also introduce a *class-state debouncer* that tracks the most recent request time for each detected object class independently. If a newly detected class has not been queried within the debounce interval, it is considered eligible for triggering VLM reasoning even

Table 1: Household hazard scenarios and their corresponding candidate objects used for detection.

Hazard Scenario	Hazard Object	Hazard Type	Candidate Object
Plush toys placed inside a baby crib	Stuffed Animal/Plush Toys	Suffocation hazard	Baby crib
Unsafe television placement	TV	Tip-over hazard	Television
Cleaning products within reach	Cleaning Products	Poison hazard	Cleaning products
Firearms accessible to children	Firearms	Injury hazard	Firearms

if another class was recently processed.

2.3.2 Scene Deduplication Middleware

Although the debouncing mechanism significantly reduces redundant requests, an additional challenge arises when the user remains stationary and continues to observe the same scene for an extended period. In such cases, the object detection module repeatedly identifies the same candidate objects, which may still trigger unnecessary VLM queries despite the debounce control. To address this issue, we introduce a perceptual hash-based scene similarity filtering mechanism, simply represented in Figure 3. For each image frame sent to the VLM server, a perceptual image hash is computed as a compact binary representation of the scene's visual structure. This hash is then compared with the hash of the previously processed frame using the Hamming distance, where a smaller distance indicates that the two frames are visually similar and a larger distance indicates greater scene change. In our implementation, we use a threshold of 15, meaning that two frames are considered sufficiently different only when more than 15 hash bits differ. This threshold was selected as a practical design choice to tolerate minor variations caused by small head movements, lighting changes, or sensor noise, while still detecting meaningful changes such as object movement, user repositioning, or the appearance of a new hazard-relevant object. If the hash distance is below this threshold, the frame is treated as visually redundant and the previous VLM result is reused, thereby reducing unnecessary inference requests and computational overhead.

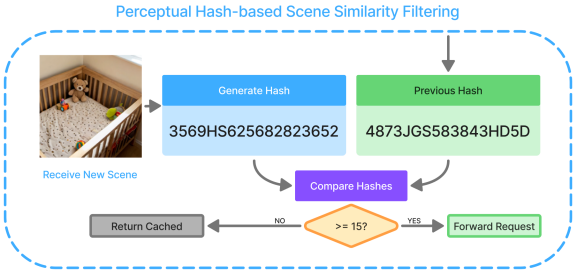


Figure 3: Perceptual hash-based scene deduplication middleware for reducing redundant VLM inference.

2.4 Affordance-Based Instruction Generation

Beyond identifying hazards, the system aims to provide actionable safety guidance for training parents, in line with guidance from major public health organizations, children's hospitals, and other injury prevention experts. To achieve this, we introduce an affordance-based instruction generation pipeline that leverages the reasoning capabilities of the VLM.

2.4.1 Safety Affordance Generation

For each detected hazard object, the VLM first generates a ranked list of potential *safety affordances*. In this context, a safety affordance refers to a safe storage location, handling mechanism, or placement option that can mitigate the hazard associated with the object. The VLM is prompted with the name of the hazardous object and instructed to produce up to six possible safe affordances.

2.4.2 Scene-Level Affordance Grounding

Once the candidate affordances are generated, the system determines whether any of these affordances are present in the current scene. To accomplish this, the captured image from the MR device is passed to the VLM along with the list of generated affordances. The VLM is then prompted to identify whether any of the listed safety affordances are visually present in the scene.

2.4.3 Instruction Generation

If an affordance is successfully grounded in the scene, the system proceeds to generate a short actionable instruction for the user. The VLM is prompted with the hazard object and the grounded affordance and instructed to produce a single concise imperative sentence describing the recommended safety action. The prompt constrains the output to a single short instruction and avoids unnecessary explanations to ensure clarity and usability. In our implementation, safety affordance generation and instruction generation are performed offline, meaning that they are precomputed before the real-time MR session rather than generated during online operation. Specifically, for each hazard category, we use the VLM server in advance to generate a set of relevant safety affordances and corresponding concise instruction templates. These precomputed affordance-instruction pairs are then stored and reused during runtime. During online operation, the system only performs scene-level affordance grounding to determine whether a relevant affordance is present in the current scene. If a grounded affordance is detected, the corresponding precomputed instruction is retrieved and displayed to the user. This design avoids invoking the VLM for instruction generation during real-time use, thereby reducing latency and computational overhead while still providing context-aware safety guidance adapted to the user's environment.

3 System Analysis

To conduct the experiment, we used the Meta Quest 3S as the HMD for the Mixed Reality client. The VLM inference server was deployed on a Dell Pro Max 16 Plus workstation equipped with 64 GB RAM and an NVIDIA RTX PRO 5000 Blackwell GPU with 24 GB VRAM. Communication between the HMD client and the VLM server was implemented using a REST API-based client-server architecture, in local network.

3.1 Hazard Inference Analysis

To evaluate the reasoning capability of the VLM for hazard detection, we created a synthetic evaluation dataset using Stable Diffusion [15]. The generated images represent different household scenarios involving hazard categories described in Tab. 1. Each image was manually categorized into two classes: **YES** (hazard present) and **NO** (hazard not present). To increase visual diversity and reduce bias, each prompt was further combined with variations in camera viewing angles and lighting conditions.

For evaluation, the VLM was prompted with a hazard-specific question corresponding to the detected object category. The model

produced a binary response indicating whether the hazardous condition was present in the scene. The predicted label was then compared with the ground-truth label obtained from the dataset. The evaluation was performed on a total of 1,025 images. The system achieved an overall accuracy of **94.73%**. The model demonstrated a high precision of **98.21%**, indicating that most scenes classified as hazardous were indeed true hazards. However, the recall of **81.41%** suggests that approximately one in five true hazards was missed, highlighting an important limitation for child-safety applications where false negatives can carry greater risk than false positives.

3.2 Real-time Performance Analysis

To evaluate the feasibility of the proposed system for real-time MR interaction, we measured the runtime performance of the complete hazard detection pipeline. To compute this metric, we repeatedly executed the complete pipeline for two candidate object classes, Cleaning Products and TV, over 39 iterations. With an model input resolution of 832×832 and passthrough camera resolution of 1280×960, the system achieved an average processing speed of approximately **67 FPS** during runtime. Across 39 test scenarios, the average end-to-end latency for a VLM response was approximately **0.987 seconds**. This latency includes image transmission, server-side inference, and response return to the MR system to display results.

4 Conclusion and Future Work

The experimental results demonstrate that the proposed OD+VLM framework can perform contextual hazard identification in real time while maintaining low latency suitable for MR applications. While the current evaluation focuses on hazard reasoning performance, and system latency, an important next step is to assess the system from the user perspective. A limitation of the current evaluation is its reliance on AI-generated synthetic images, future work will address this by incorporating real-world MR scene data collected from user study sessions.

In future work, we plan to conduct a within-subject user study to evaluate the usability, feasibility, and overall effectiveness of the proposed MR-based parent training system. The study will compare three conditions: a baseline text-only AR instruction system, a VLM-only pipeline that performs both hazard detection and reasoning, and the proposed OD+VLM pipeline. Each participant will complete randomized and counterbalanced hazard-resolution tasks across the three conditions to reduce ordering and learning effects. After each condition, participants will complete standardized post-condition surveys to evaluate perceived usability and cognitive workload. The study will also examine operational efficiency through task completion performance and hazard recognition accuracy. We expect this evaluation to provide important insights into how contextual hazard feedback influences user interaction, decision making, and safety behavior in MR environments. These findings will guide future improvements toward more human-centered, reliable, and deployable MR safety training systems.

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